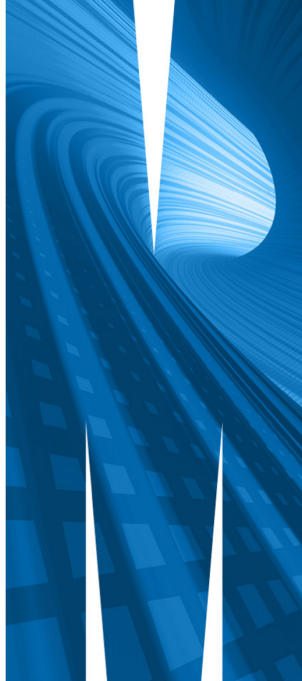


ETX1100/ETX5900

Business Statistics

Week 2: Probability and distributions

1 December 2026



Outline

- 1 Probability language
- 2 Contingency tables
- 3 Continuous probability distributions
- 4 Integrated application

Today's journey

- 1 Translate business uncertainty into events and probabilities.
- 2 Build contingency tables and distinguish joint, marginal, and conditional probability.
- 3 Assess whether two events are independent.
- 4 Use Normal and Student's t distributions in Excel.
- 5 Finish with an integrated marketing-and-risk case.

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Retrieval check

- Is Have Tried categorical or numerical?
- Is Income categorical or numerical?
- What does one row of the Marketing worksheet represent?
- Why should a business question come before an Excel calculation?

Probability measures uncertainty

- An **experiment** is a process with an uncertain outcome.
- The **sample space** S contains all possible outcomes.
- An **event** is one or more outcomes of interest.
- $0 \leq P(A) \leq 1$ and $P(S) = 1$.
- Probabilities can be expressed as proportions or percentages.

AND, OR, and NOT

Business wording	Notation	Meaning
A and B	$A \cap B$	both occur
A or B	$A \cup B$	at least one occurs
not A	A^c	complement of A

$$P(A^c) = 1 - P(A)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Mutually exclusive is not independent

- **Mutually exclusive:** A and B cannot occur together, so $P(A \cap B) = 0$.
- **Independent:** learning that B occurred does not change the probability of A .
- Non-trivial mutually exclusive events are dependent: observing one rules out the other.

Watch me: translate a business question

For an online order, define:

- A : delivery is late;
- B : customer requests a refund.

Translate:

- 1 “late and refunded” as $A \cap B$;
- 2 “late or refunded” as $A \cup B$;
- 3 “not late” as A^c ; and
- 4 “refunded given late” as $P(B \mid A)$.

Now you try: events and rules

Time allowed: 4 minutes

Let C mean a customer clicks an advertisement and P mean they purchase.

- 1 Translate “clicked but did not purchase”.
- 2 Translate “clicked or purchased”.
- 3 If $P(C) = 0.30$, $P(P) = 0.12$, and $P(C \cap P) = 0.08$, calculate $P(C \cup P)$.
- 4 Explain why click and purchase are not mutually exclusive.

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Two categorical variables

A contingency table cross-classifies observations by two categorical variables.

- **Cell:** joint frequency or probability.
- **Margin:** total for one variable.
- **Grand total:** all observations.
- Percent of grand total supports joint and marginal probability.
- Row or column percentages support conditional probability.

Watch me: build the Marketing table

Open week2-categorical-data.xlsx and select **Marketing**.

Create a PivotTable:

- 1 put Income in Rows;
- 2 put Have Tried in Columns;
- 3 count Person ID in Values; and
- 4 show the counts as **% of Grand Total**.

Check that all cells sum to 100%.

Joint and marginal probability

- A **joint probability** refers to two events together: $P(A \cap B)$.
- A **marginal probability** refers to one variable and is read from a row or column total.

Always state the denominator:

$$P(A) = \frac{\text{observations satisfying } A}{\text{all observations}}.$$

Now you try: marketing probabilities

Time allowed: 7 minutes

Using the PivotTable, calculate and label:

- 1 $P(\text{not tried})$;
- 2 $P(\text{income} > 50,000 \cap \text{not tried})$;
- 3 $P(\text{income} > 50,000 \cup \text{not tried})$.

For each answer, identify whether it is marginal, joint, or union probability.

Conditional probability

Conditional probability changes the denominator:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

Read $P(A \mid B)$ as “the probability of A , given B ”. The event after the vertical bar defines the group being considered.

Watch me: condition on “not tried”

To calculate

$$P(\text{income} > 50,000 \mid \text{not tried}),$$

either:

- divide the relevant joint probability by $P(\text{not tried})$; or
- show PivotTable values as **% of Column Total** if Have Tried is in columns.

Verify that each conditioning column sums to 100%.

Now you try: reverse the condition

Time allowed: 6 minutes

Calculate both:

1 $P(\text{not tried} \mid \text{income} > 50,000);$

2 $P(\text{income} > 50,000 \mid \text{not tried}).$

Explain why these are different questions. Which denominator is used in each calculation?

Independence

Events A and B are independent when any equivalent condition holds:

$$P(A \mid B) = P(A),$$

$$P(B \mid A) = P(B),$$

$$P(A \cap B) = P(A)P(B).$$

A small difference may arise from sampling variation; it is not

Watch me: tried product and gender

Create a second PivotTable for Gender and Have Tried.

Compare:

$$P(\text{tried} \mid \text{female}) \quad \text{with} \quad P(\text{tried}).$$

Narrate the conclusion as an association in this sample, not a causal effect.

Now you try: assess independence

Time allowed: 7 minutes

Use Live Alone and Have Tried.

- 1 Construct the contingency table.
- 2 Compare an appropriate conditional probability with its marginal probability.
- 3 Decide whether independence looks plausible.
- 4 Write one sentence that avoids causal language.

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From data to a probability model

- A histogram describes observed data.
- A probability density curve models possible values of a continuous variable.
- Probability is represented by **area under the curve**.
- Total area is 1.
- For a continuous variable, probability at one exact value is effectively 0.

The Normal distribution

If $X \sim N(\mu, \sigma)$, the distribution is:

- symmetric and bell-shaped;
- centred at mean μ ; and
- spread according to standard deviation σ .

Approximately 68%, 95%, and 99.7% lie within one, two, and three standard deviations of the mean.

Excel's Normal functions

Question	Excel function
$P(X \leq x)$	NORM.DIST(x, mean, sd
$P(X > x)$	1-NORM.DIST(x, mean,
percentile with lower-tail probability p	NORM.INV(p, mean, sd)
standard Normal percentile	NORM.S.INV(p)

For an interval, subtract two cumulative probabilities.

Watch me: blood pressure risk

Assume adult systolic blood pressure is Normal with mean 124 and standard deviation 10.

- Below 117: `=NORM.DIST(117,124,10,TRUE)`
- Above 140: `=1-NORM.DIST(140,124,10,TRUE)`
- Highest 15% cutoff: `=NORM.INV(0.85,124,10)`

Sketch and shade the requested area before typing the formula.

Now you try: service time

Time allowed: 7 minutes

Assume service time is Normal with mean 8 minutes and standard deviation 1.5 minutes.

- 1 Find $P(X < 6)$.
- 2 Find $P(X > 10)$.
- 3 Find $P(7 < X < 9)$.
- 4 Find the 90th percentile.
- 5 For each, sketch the relevant area and write the Excel formula.

Standardising a value

The standard score

$$z = \frac{X - \mu}{\sigma}$$

counts how many standard deviations x is above or below the mean.

- $z > 0$: above the mean.
- $z < 0$: below the mean.
- Magnitude describes distance from the mean in standard-deviation units.

Now you try: compare unlike measures

Time allowed: 5 minutes

A store's weekly sales are 1.4 standard deviations above the chain mean. Its profit is 0.6 standard deviations above the chain mean.

- 1 Which result is more unusual relative to its distribution?
- 2 Why can standard scores compare variables measured in different units?
- 3 Does the comparison prove the store is well managed?

Student's t distribution

- The t distribution is symmetric and centred at zero.
- It has heavier tails than the standard Normal distribution.
- Its shape depends on degrees of freedom, commonly $df = n - 1$.
- It is used when estimating a mean and the population standard deviation is unknown.
- As df increases, t approaches the standard Normal distribution.

Watch me: choose the distribution

Ask in order:

- 1 Are we modelling individual values or a sample statistic?
- 2 Is a Normal model for the population stated or defensible?
- 3 Is the population standard deviation known?
- 4 If estimating a mean with unknown σ , what are the degrees of freedom?

The method follows the information available—not the desired answer.

Now you try: method selection

Time allowed: 5 minutes

Choose Normal, standard Normal, t , or “not enough information”:

- 1 Find a percentile of stated Normal delivery times.
- 2 Standardise one delivery time when μ and σ are known.
- 3 Estimate a population mean from $n = 18$ observations when σ is unknown.
- 4 Model a severely skewed individual outcome with no distributional assumption.

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Watch me: from question to conclusion

For “Are high-income respondents more likely to have tried the product?”:

- 1 define the two events;
- 2 build the correct PivotTable;
- 3 choose conditional probabilities with clear denominators;
- 4 compare them; and
- 5 state the sample association and its limitation.

The conclusion should answer the business question, not narrate menu clicks.

Now you try: marketing evidence brief

Time allowed: 10 minutes

Using the Marketing worksheet, choose one demographic variable and answer:

- 1 Is product trial associated with that variable?
- 2 Which joint, marginal, and conditional probabilities support your answer?
- 3 Is independence plausible?
- 4 What can and cannot be concluded from these observational data?

Prepare a three-sentence briefing and one clearly labelled

Common traps

- Treating “and” as addition without subtracting overlap.
- Reversing $P(A \mid B)$ and $P(B \mid A)$.
- Comparing conditional probabilities with different, hidden denominators.
- Calling mutually exclusive events independent.
- Using `NORM.DIST` without checking the tail or interval requested.
- Treating a model assumption as an empirical fact.

Closing check

Can you:

- translate business language into event notation?
- distinguish joint, marginal, union, and conditional probability?
- test the idea of independence using a contingency table?
- choose the correct Excel Normal function and tail?
- explain when a t distribution becomes relevant?
- communicate association without claiming causation?